

Remote App Update Integration Guide

Target: Flutter Mobile Developer | Status: Live on Dev Backend

Core Principle: The backend owns the update policy rules; Flutter enforces them. No minimum versions are hard-coded in Flutter. Admins control rollout and enforcement remotely from the admin dashboard.

1. Public API Endpoint (No Auth Required)

Call this endpoint during Flutter app startup and when returning to foreground:

```
GET /api/v1/app/update-config?platform=android&version=2.4.0
```

Request Parameters:

Parameter	Type	Required	Description
platform	string	Yes	Client platform: android or ios (lowercase)
version	string	Yes	Installed app SemVer version (e.g. 2.4.0)
appld	string	No	App flavor or identifier (e.g. com.oreedu.app)

2. JSON Response Contract

```
{
  "platform": "android",
  "currentVersion": "2.4.0",
  "minimumSupportedVersion": "2.5.0",
  "latestVersion": "2.6.0",
  "updateType": "force",
  "updateRequired": true,
  "title": "Update Required",
  "message": "A new version of the app is required to continue using the service.",
  "storeUrl": "https://play.google.com/store/apps/details?id=com.oreedu.app",
  "policyVersion": 2,
  "publishedAt": "2026-09-14T05:15:00.000Z"
}
```

Understanding updateType Actions:

updateType	updateRequired	Flutter Client Action
force	true	Block user with ForceUpdateScreen. Back button disabled. Only "Update Now" action.
optional	false	Show optional update dialog ("Update" / "Later"). Allow user to continue.
none	false	Installed version is up to date. Open normal app flow directly.

3. Flutter Client Implementation Requirements

- Root Startup Gate: Check the endpoint before showing login or loading authenticated data.
- Back Button Disabled: Wrap ForceUpdateScreen in PopScope(canPop: false) to prevent bypassing.
- App Resume Re-check: Listen to AppLifecycleState.resumed to re-verify version when user returns from App Store.
- Fail-Open Fallback: Set a 5-second timeout. If network fails or offline, allow app entry without blocking.
- Store Redirect: Launch storeUrl using url_launcher with LaunchMode.externalApplication.

4. Admin Management Portal

- Accessible under Platform Settings -> App Updates (/mobile-updates).
- Independent version management for Android and iOS.
- Interactive live version tester with phone mockup simulator.
- Safety confirmation prompt before enabling Force Updates.